

FLEET RULE OF THE ROAD TEST FOR SEP 2017

This test is based on: The International Regulations for Preventing Collisions at Sea (1972) with amendments in force up to December 2009 as detailed in "A Seaman's Guide Pocket Book of The International Regulations for Preventing Collisions at Sea" (BR 453 Supp), ATP1(G) Vol II, NP735 and the Mariners Handbook (NP100).

The test is designed to be given to any Warfare Officer, Deck officer, Officer undertaking bridge watch-keeping duties or Young Officer/Cadet. The test should be administered under written exam conditions, the results should be recorded in the OOW training records for inspection by FOST navigation staff. The exam should take no longer than 1hr.

In this test, any change in the status of the ship or of the environment will be indicated in the script. Rules 1 to 19, 34, 35 and Annex IV must be known instinctively for instant application. A thorough knowledge of the other rules is required. The pass mark is 90%.

PART 1

1. Describe a westerly cardinal mark, including colour, shape, topmark and light characteristics. **(NP375)(2)**.

2. IRPCS Rule 15 does not apply between screen ships and ships of the main body for Allied naval vessels. True or False? **(ATP 1G Vol II Art 140b)(0.5)**

3. When a formation consists of a main body and screen, which group has the right of way (ships of the screen or of the main body)?**(ATP 1G Vol II Art 140b)(0.5)**

4. If the vessel required to maintain her course and speed finds herself so close that collision cannot be avoided by the action of the give way vessel alone, what action shall she take?**(ATP 1G Vol II Art 140b)(0.5)**

5. The Rules shall not interfere with the operation of special rules made by whom and in what areas? **(R1b)(1)**

6. To what should due regard be had when construing and complying with these Rules?(R2b)(1)
7. Define the word "vessel"?(R3a)(1)
8. The term restricted visibility means what?(R3l)(1)
9. When and how **may** a vessel required to keep her course and speed take action to avoid collision?(R17a(ii))(1)
10. Rule 8f(i) & 8f(ii) mention vessels that are *not to impede the passage or safe passage of another vessel*. Name three such vessels with the associated circumstances. (R9 b, c & d, R10i, j and R18di)(3)
11. When required by the circumstances of the case, how should a vessel required not to impede the passage or safe passage of another vessel act? (R8f(i))(1)
12. Describe 4 factors that **all vessels** should take into account when determining safe speed.(R6a)(2)

13. The Rules regarding restricted visibility applies to vessels **in sight of one another** when navigating in or near an area of restricted visibility; True or False? **(R19a)(0.5)**

14. Describe the actions required by the Rules for a vessel which detects by radar alone the presence of another vessel. Include actions that should be avoided so far as possible?**(R19d)(3)**

15. Describe the lights displayed by a vessel engaged in trawling (greater than 20m in length) when hauling her nets in close proximity to other vessels engaged in fishing?**(R26 b(i), d and annex II)(2)**

Part 1

_____/20

Part 2

You are the Navigating Officer of HMS TYNE (length 79m) on a Western Patrol. DEFRA has declared a Spanish Trawler (length 55m) as a contact of interest (COI) and has tasked you to the "Trouser leg" to board and investigate. The UK is sitting under the influence of a high pressure which is giving both fine weather and foggy periods. The CO decides to approach covertly - under the cover of darkness, using deceptive lighting and maintaining Radio silence.

16. Justify your recommendation to the CO on the use of deceptive lighting?(**R1c**)(1)

17. During the transit, your lookout reports a merchant vessel at Red 160 at 3nm. You determine she is drawing left with a CPA of 1c on your starboard quarter. What is the responsibility of this vessel if a close quarters situation is deemed to be developing?(**R13a**)(1)

18. The merchant vessel passes and opens up to 3nm and now at Green 30, she alters course to port unexpectedly, possibly to proceed into port, and is on a steady bearing in 10 minutes. Who is the give way vessel noting the previous situation in the question above? (**R13d**)(1)

19. Both vessels continue to close on a steady bearing. When and how **shall** the stand-on vessel take action to avoid collision?(**17b**)(1)

20. How does the Rule require the vessel directed to keep out of the way of another vessel to act?(**R16**)(1)

21. The situation is resolved eventually. The latest report on the COI indicates she is within 10nm of your current position. What lights will you brief your lookouts to look for if you consider her to be trawling and making way?(**R26b**)(2)

22. The CO overhears your brief to the lookouts and asks what other possible lights could she be displaying if she is not engaged in fishing?(**R26e, 23a**)(2)

23. You positively identify the COI after sunset and the CO decides to board at first light. In the meantime you are to remain over the horizon and maintain radar contact. What shapes will you expect the COI to display during the day? (**R26b(i)**)(1)

24. The boarding goes ahead as intended and the Boarding Officer report a couple of infringements which leads to the vessel being detained. In consultation with DEFRA, the decision is made to take the detained vessel to Milford Haven. You embark a steaming party on the COI for the transit. All seem to be going well until the embarked team report the vessel has lost all propulsion and not under command. What shapes will you expect them to display?(**R27a**)(1)

25. What lights would this be at night if they were making way through the water. How will this change when they eventually stop in the water? (**R27a**)(3)

26. After several phone calls to the various stakeholders and a risk assessment, it is decided to conduct a non-contractual tow. What shape(s), if any, will both vessels display if the length of the tow is less than 200m?(**R24a(v) & e(iii)**)(1)

27. As you are not normally engaged in towing operations and therefore unable to display the lights prescribed for towing, what additional measure(s) does the Rules require you to take?(**R24i**)(1)

28. You successfully take the COI in tow and begin your transit to Milford Haven. During the latter half of the First Watch, you start experiencing restricted visibility. Describe the sound signals for you (the towing vessel) and the towed vessel. **(R35 c&e)(3)**

29. Your look out reports a sound signal forward of the ship at medium range consisting of one prolonged followed by two short blasts at intervals not more than 2 minutes. Describe your initial actions in accordance with the Rules. **(R19e)(1)**

30. Having taken the appropriate action, you subsequently detect on radar a contact at Green 20 at 3.2nm on ARPA. The ARPA is displaying between a steady and 5c CPA. What warning does the Rule give regarding the use of such ARPA information? **(R7c)(1)**

31. You later sight a power driven vessel at Green 20 at a range of 1000yds. You are 80 degrees on her port bow and she is drawing left with a CPA of 3c ahead of the towing group. Who is the give way vessel (you do not consider yourself to be restricted in your ability to follow the course you and the tow are following)? **(R15)(1)**

32. You decide to anchor in the Milford Haven anchorage until the visibility clears prior to entering harbour. You have tugs on stand-by to assist with anchoring the COI. What sound signals will you and the COI make at anchor? **(R35g)(1)**

33. What lights and/or shapes will both vessels display whilst at anchor? **(R30 a &c)(3)**

34. Milford Haven Port control decides to send a Pilot in addition to the Tugs to assist in anchoring the COI. How will you identify the Pilot vessel in the prevailing conditions of restricted visibility?(R29a(i)(ii), R35 a, b & k)(4)

Both you and the COI anchor safely and you begin a waiting game for the visibility to clear.

Part 2 _____/30

TOTAL _____/50

PART 1

1. Describe a westerly cardinal mark, including colour, shape, topmark and light characteristics. **(NP375)(2)**.

Colour. Yellow with a single broad horizontal black band

Shape. Pillar or Spar

Topmark. 2 black cones, one above the other

Light. White, VQ(9) 10s or Q(9) 15s

2. IRPCS Rule 15 does not apply to between screen ships and ships of the main body for Allied naval vessels. True or False? **(ATP 1G Vol II Art 140b)(0.5)**

True

ATP 1G Vol II Art 140b. Screen ships, when a formation consists of a main body and screen, ships of the screen are to keep clear of the those of the main body. When, from any course, a ship of the main body finds herself so close that collision cannot be avoided by action of the screen ship alone, she shall take such action as will best avoid collision. Rule 15 of the IRPCS is not to apply between screen ships and ships of the main body.

3. When a formation consists of a main body and screen, which group has the right of way (ships of the screen or of the main body)?**(ATP 1G Vol II Art 140b)(0.5)**

See above

4. If the vessel required to maintain her course and speed finds herself so close that collision cannot be avoided by the action of the give way vessel alone, what action shall she take?**(ATP 1G Vol II Art 140b)(0.5)**

See above

5. The Rules shall not interfere with the operation of special rules by whom and in what areas? **(R1b)(1)**

R1b. Nothing in these Rules shall interfere with the operation of special rules made by an appropriate authority for roadsteads, harbours, rivers, lakes or inland waterways connected with the high seas and navigable by seagoing vessels. Such special rules shall conform as closely as possible to these Rules.

6. To what should due regard be had when construing and complying with these Rules?**(R2b)(1)**

R2b. In construing and complying with these Rules due regard shall be had to all dangers of navigation and collision and to any special circumstances, including the limitations of the vessels involved, which may make a departure from these Rules necessary to avoid immediate danger

7. Define the word "vessel"?**(R3a)(1)**

R3a. The word “vessel” includes every description of water craft, including non-displacement craft, WIG craft and seaplanes, used or capable of being used as a means of transportation on water.

8. The term restricted visibility means what? **(R3l)(1)**

R3l. The term “restricted visibility” means any condition in which visibility is restricted by fog, mist, falling snow, heavy rainstorms, sandstorms or any other similar causes

9. When and how **may** a vessel required to keep her course and speed take action to avoid collision? **(R17a(ii))(1)**

R17a(ii). The latter vessel may however take action to avoid collision by her manoeuvre alone, as soon as it becomes apparent to her that the vessel required to keep out of the way is not taking appropriate action in compliance with these Rules.

10. Rule 8f(i) & 8f(ii) mention vessels that are *not to impede the passage or safe passage of another vessel*. Name three such vessels and circumstances. **(R9 b, c & d, R10i, j and R18di)(R3)**

R9b. A vessel of less than 20 metres in length or a sailing vessel shall not Impede the passage of a vessel which can safely navigate only within a Narrow Channel or Fairway.

R9c. A vessel Engaged in Fishing shall not Impede the passage of any other vessel navigating within a Narrow Channel or Fairway.

R9d. A vessel shall not cross a Narrow Channel or Fairway if such Crossing Impedes the passage of a vessel which can safely navigate only within such channel or Fairway.

R10i. A vessel Engaged in Fishing shall not Impede the passage of any vessel following a Traffic Lane.

R10j. A vessel of less than 20 metres in length or a sailing vessel shall not Impede the safe passage of a power-driven vessel following a Traffic Lane.

R18di. Any vessel other than a vessel Not Under Command or a vessel Restricted in her Ability to Manoeuvre shall, if the circumstances of the case admit, avoid Impeding the safe passage of a vessel Constrained by her Draught, exhibiting the signals in Rule 28.

11. When required by the circumstances of the case, how should a vessel required not to impede the passage or safe passage of another vessel act? **(R8f(i))(1)**

R8f(i). A vessel which, by any of these Rules, is required not to impede the passage or safe passage of another vessel shall, when required by the circumstances of the case, take early action to allow sufficient sea-room for the safe passage of the other vessel.

12. Describe 4 factors that **all vessels** should take into account when determining safe speed. **(R6a)(2)**

R6a. In determining a safe speed the following factors shall be among those taken into account:

(a). By all vessels:

(i). the state of visibility;

(ii). the traffic density including concentrations of fishing vessels or any other vessels;

(iii). the manoeuvrability of the vessel with special reference to stopping distance and turning ability in the prevailing conditions;

(iv). at night the presence of background light such as from shore lights or from back scatter of her own lights;

(v). the state of wind, sea and current, and the proximity of navigational hazards;

(vi). the draught in relation to the available depth of water

13. The Rules regarding restricted visibility applies to vessels **in sight of one another** when navigating in or near an area of restricted visibility; True or False? **(R19a)(0.5)**

False

R19a. This Rule applies to vessels not in sight of one another when navigating in or near an area of restricted visibility

14. Describe the actions required by the Rules for a vessel which detects by radar alone the presence of another vessel. Include actions that should be avoided as far as possible? **(R19d)(3)**

R19d. A vessel which detects by radar alone the presence of another vessel shall determine if a close-quarters situation is developing and/or risk of collision exists. If so, she shall take avoiding action in ample time, provided that when such action consists of an alteration of course, so far as possible the following shall be avoided:

(i). an alteration of course to port for a vessel forward of the beam, other than for a vessel being overtaken;

(ii). an alteration of course towards a vessel abeam or abaft the beam.

15. Describe the lights displayed by a vessel engaged in trawling (greater than 20m in length) when hauling her nets in close proximity to other vessels engaged in fishing? **(R26 b(i), d and annex II)(2)**

R26b(i). A vessel when engaged in trawling, by which is meant the dragging through the water of a dredge net or other apparatus used as a fishing appliance, shall exhibit:

(i). two all-round lights in a vertical line, the upper being green and the lower white, or a shape consisting of two cones with their apexes together in a vertical line one above the other;

R26d. The additional signals described in Annex II to these regulations apply to a vessel engaged in fishing in close proximity to other vessels engaged in fishing

Annex II Vessels of 20 metres or more in length when engaged in trawling, whether using demersal or pelagic gear, shall exhibit:

(ii). when hauling their nets:

one white light over one red light in a vertical line;

Part 1

_____/20

Part 2

You are the Navigating Officer of HMS TYNE (length 79m) on a Western Patrol. DEFRA has declared a Spanish Trawler (length 55m) as a contact of interest (COI) and has tasked you to the "Trouser leg" to board and investigate. The UK is sitting under the influence of a high pressure giving both fine weather and foggy periods. The CO decides to approach covertly - under the cover of darkness using deceptive lighting and maintaining Radio silence.

16. Justify your recommendation to the CO on the use of deceptive lighting?(**R1c**)(1)

R1c. Nothing in these Rules shall interfere with the operation of any special rules made by the Government of any State with respect to additional station or signal lights, shapes or whistle signals for ships of war and vessels proceeding under convoy, or with respect to additional station or signal lights or shapes for fishing vessels engaged in fishing as a fleet. These additional station or signal lights, shapes or whistle signals shall, so far as possible, be such that they cannot be mistaken for any light, shape or signal authorized elsewhere under these Rules

17. During the transit, your lookout reports a merchant vessel at Red 160 at 3nm. You determine she is drawing left with a CPA of 1c on your starboard quarter. What is the responsibility of this vessel if a close quarters situation is deemed to be developing?(**R13a**)(1)

R13a. Notwithstanding anything contained in the Rules of part B, sections I and II, any vessel overtaking any other shall keep out of the way of the vessel being overtaken

18. As the merchant vessel passes and opens up to 3nm at Green 30, she alters course to port unexpectedly, possibly to proceed into port, and is on a steady bearing in 10 minutes. Who is the give way vessel noting the previous situation in the question above? (**R13d**)(1)

R13d. Any subsequent alteration of the bearing between the two vessels shall not make the overtaking vessel a crossing vessel within the meaning of these Rules or relieve her of the duty of keeping clear of the overtaken vessel until she is finally past and clear

19. Both vessels continue to close on a steady bearing. When and how **shall** the stand-on vessel take action to avoid collision?(**17b**)(1)

R17b. When, from any cause, the vessel required to keep her course and speed finds herself so close that collision cannot be avoided by the action of the give-way vessel alone, she shall take such action as will best aid to avoid collision

20. How does the Rule require the vessel directed to keep out of the way of another vessel to act?(**R16**)(1)

R16. Every vessel which is directed to keep out of the way of another vessel shall, so far as possible, take early and substantial action to keep well clear.

21. The situation is resolved eventually. The latest report on the COI indicates she is within 10nm of your current position. What lights will you brief your lookouts to look for, if you consider her to be trawling and making way?(**R26b**)(2)

R26b. A vessel when engaged in trawling, by which is meant the dragging through the water of a dredge net or other apparatus used as a fishing appliance, shall exhibit:

- (i). two all-round lights in a vertical line, the upper being green and the lower white, or a shape consisting of two cones with their apexes together in a vertical line one above the other;*
- (ii). a masthead light abaft of and higher than the all-round green light; a vessel of less than 50 metres in length shall not be obliged to exhibit such a light but may do so;*
- (iii). when making way through the water, in addition to the lights prescribed in this paragraph, sidelights and a sternlight.*

22. The CO overhears your brief to the lookouts and asks what other possible lights could she be displaying if she were not engaged in fishing?(**R26e, 23a**)(2)

R23a. A power-driven vessel underway shall exhibit:

- (i). a masthead light forward;*
- (ii). a second masthead light abaft of and higher than the forward one; except that a vessel of less than 50 metres in length shall not be obliged to exhibit such light but may do so;*
- (iii). sidelights;*
- (iv). a sternlight.*

23. You positively identify the COI and the CO decides to board at first light. In the meantime you are to remain over the horizon and maintain radar contact. What shapes will you expect the COI to display during the day? (**R26b(i)**)(1)

R26b(i). ...two all-round lights in a vertical line, the upper being green and the lower white, or a shape consisting of two cones with their apexes together in a vertical line one above the other

24. The boarding goes ahead as intended and the Boarding Officer report a couple of infringements which leads to the vessel being detained. In consultation with DEFRA, the decision is made to take the detained vessel to Milford Haven. You embark a steaming party on the COI for the transit. All seem to be going well until the embarked team report the vessel has lost all propulsion and not under command. What shapes will you expect them to display?(**R27a**)(1)

R27a. A vessel not under command shall exhibit:

- (i). two all-round red lights in a vertical line where they can best be seen;*
- (ii). two balls or similar shapes in a vertical line where they can best be seen;*

25. What lights would this be at night if they were making way through the water. How will this change when they eventually stop in the water? (**R27a**)(3)

R27a. A vessel not under command shall exhibit:

- (i). two all-round red lights in a vertical line where they can best be seen;*
- (ii). two balls or similar shapes in a vertical line where they can best be seen;*

(iii). when making way through the water, in addition to the lights prescribed in this paragraph, sidelights and a sternlight

26. After several phone calls to the various stakeholders and a risk assessment, It is decided to conduct a non-contractual tow. What shape(s), if any, will both vessels display if the length of the tow is less than 200m?(R24a(v) & e(iii)(1)

None required.

R24a(v). when the length of the tow exceeds 200 metres, a diamond shape where it can best be seen.

R24e(iii). when the length of the tow exceeds 200 metres, a diamond shape where it can best be seen

27. As you are not normally engaged in towing operations and therefore unable to display the lights prescribed for towing, what additional measure(s) does the Rules require you to take?(R24i)(1)

R24i. Where from any sufficient cause it is impracticable for a vessel not normally engaged in towing operations to display the lights prescribed in paragraph (a) or (c) of this Rule, such vessel shall not be required to exhibit those lights when engaged in towing another vessel in distress or otherwise in need of assistance. All possible measures shall be taken to indicate the nature of the relationship between the towing vessel and the vessel being towed as authorized by Rule 36, in particular by illuminating the towline

28. You successfully take the COI in tow and begin your transit to Milford Haven. During the latter half of the First Watch, you start experiencing restricted visibility. Describe the sound signals for you (the towing vessel) and the towed vessel.(R35 c&e)(3)

R35c. A vessel not under command, a vessel restricted in her ability to manoeuvre, a vessel constrained by her draught, a sailing vessel, a vessel engaged in fishing and a vessel engaged in towing or pushing another vessel shall, instead of the signals prescribed in paragraphs (a) or (b) of this Rule, sound at intervals of not more than 2 minutes three blasts in succession, namely one prolonged followed by two short blasts

R35e. A vessel towed or if more than one vessel is towed the last vessel of the tow, if manned, shall at intervals of not more than 2 minutes sound four blasts in succession, namely one prolonged followed by three short blasts. When practicable, this signal shall be made immediately after the signal made by the towing vessel

29. Your look out reports a sound signal forward of the ship at medium range consisting of one prolonged followed by two short blasts at intervals not more than 2 minutes. Describe your initial actions in accordance with the Rules. (R19e)(1)

R19e. Except where it has been determined that a risk of collision does not exist, every vessel which hears apparently forward of her beam the fog signal of another vessel, or which cannot avoid a close-quarters situation with another vessel forward of her beam, shall reduce her speed to the minimum at which she can be kept on her course. She shall if necessary take all her way off and in any event navigate with extreme caution until danger of collision is over

30. Having taken the appropriate action, you subsequently detect on radar a contact at Green 20 at 3.2nm on ARPA. The ARPA is displaying between a steady and 5c CPA. What warning does the Rule give regarding the use of such ARPA information?(R7c)(1)

R7c. Assumptions shall not be made on the basis of scanty information, especially scanty radar information

31. You finally sight a power driven vessel at Green 20 at a range of 1000yds. You are 80 degrees on her port bow and she is drawing left with a CPA of 3c ahead of the towing group. Who is the give way vessel, if you do not consider yourself to be restricted in your ability to follow the course you and the tow are following?(R15)(1)

R15. When two power-driven vessels are crossing so as to involve risk of collision, the vessel which has the other on her own starboard side shall keep out of the way and shall, if the circumstances of the case admit, avoid crossing ahead of the other vessel

32. You decide to anchor in the Milford Haven anchorage until the visibility clears prior to entering harbour. You have tugs on stand to assist with anchoring the COI. What sound signals will you and the COI make at anchor?(R35g)(1)

R35g. A vessel at anchor shall at intervals of not more than one minute ring the bell rapidly for about 5 seconds. In a vessel of 100 metres or more in length the bell shall be sounded in the forepart of the vessel and immediately after the ringing of the bell the gong shall be sounded rapidly for about 5 seconds in the after part of the vessel. A vessel at anchor may in addition sound three blasts in succession, namely one short, one prolonged and one short blast, to give warning of her position and of the possibility of collision to an approaching vessel

33. What lights and/or shapes will both vessels display whilst at anchor? (R30 a &c)(3)

R30a. A vessel at anchor shall exhibit where it can best be seen:

(i). in the fore part, an all-round white light or one ball;

(ii). at or near the stern and at a lower level than the light prescribed in subparagraph (i), an all-round white light

R30c. A vessel at anchor may, and a vessel of 100 metres and more in length, shall also use the available working or equivalent lights to illuminate her decks

34. Milford Haven Port control decides to send a Pilot in addition to the Tugs to assist in anchoring the COI. How will you identify the Pilot vessel in the prevailing conditions of restricted visibility?(R29a(i)(ii), R35 a, b & k)(4)

R29a. A vessel engaged on pilotage duty shall exhibit:

(i). at or near the masthead, two all-round lights in a vertical line, the upper being white and the lower red;

(ii). when underway, in addition, sidelights and a sternlight;

R35

- a. A power-driven vessel making way through the water shall sound at intervals of not more than 2 minutes one prolonged blast.
- b. A power-driven vessel underway but stopped and making no way through the water shall sound at intervals of not more than 2 minutes two prolonged blasts in succession with an interval of about 2 seconds between them.
- k. A pilot vessel when engaged on pilotage duty may in addition to the signals prescribed in paragraphs (a), (b) or (g) of this Rule sound an identity signal consisting of four short blasts

Both you and the COI anchor safely and you begin a waiting game for the visibility to clear.

Part 2 _____/30

TOTAL _____/50